Angle bisectors and perpendicular bisectors

Core Concepts: Angle bisectors and perpendicular bisectors

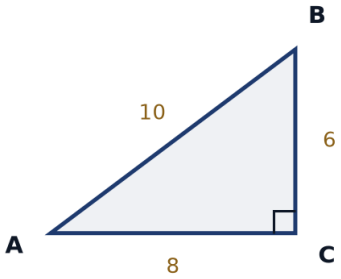
1 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 94°

b. 104°

c. 86°

d. 139°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 8

c. 6

d. 8

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 139°

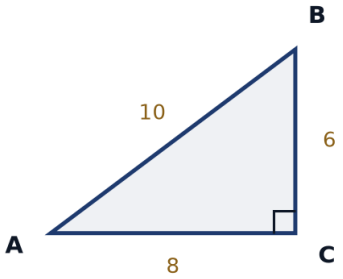
5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm

c. 20 cm

d. 45 cm



6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 6

b. 10

c. 8

d. 10

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 80°

b. 90°

c. 72°

d. 139°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

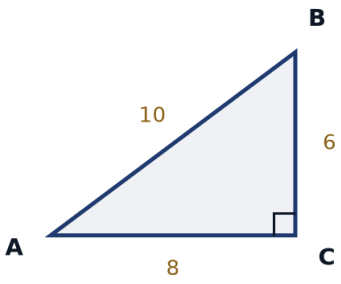
9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 8

b. 12

c. 10

d. 12



Exam-Style Problem Solving & Modeling

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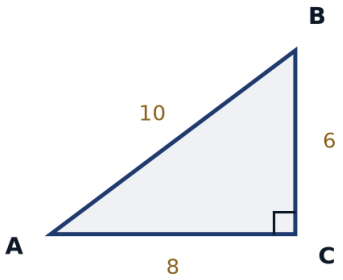
10 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 73°

b. 83°

c. 65°

d. 139°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 10

b. 14

c. 12

d. 14

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